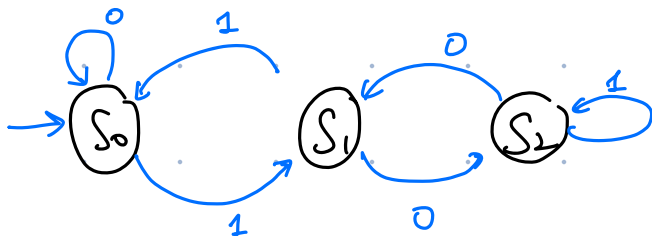


Morning Quiz



My guess:

$$\begin{array}{r} 101 \\ 421 \\ \hline \end{array} = 5$$

$$\begin{array}{r} 1011 \\ 8421 \\ \hline \end{array} = 8+3 = 11$$

} this is not the right way to think about it.

My goal is to start at S_0 and come back to S_0 .

within
0000
+0
1111

$$① 0 \Rightarrow 0$$

$$② 0011 \Rightarrow 3$$

$$③ 0110 \Rightarrow 6$$

$$④ \begin{array}{r} 1001 \\ 8421 \\ \hline \end{array} \Rightarrow 9$$

} multiples of 3.

Why is this multiple of 3?

Any number mod 3 have

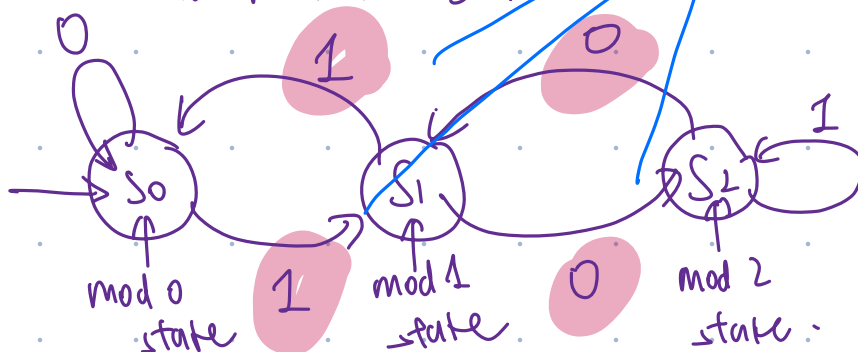
3 cases

$$N \bmod 3 \equiv 0$$

$$N \bmod 3 \equiv 1$$

$$N \bmod 3 \equiv 2$$

shows how you can transition between different classes.



→ A grammar for a (natural) language precisely specifies which sentences are grammatical.

→ There are 2 ways of understanding a formal language description

→ classifies which strings are in the language.

→ Generate the strings in this sentence

Claude Shannon: The Redundancy of English.

↳ Presents some statistical models to generate English.

① uniform at random (characters)

② letter randomly sampled w/ probability they appear in normal English. (unigram)

③ choose 1 letter | choose next letter based on prev letter

(bigram & trigrams)

④ Unigram ~~with~~ words frequencies.

⑤ Bigram word frequencies.

they are all finite state automata!

Natural languages:

N-grams: as N increases, do we converge to English?

Three models for the description of languages.

1 $\hookrightarrow$ Chomsky's Paper

$\hookrightarrow$ Context-free grammars

$\downarrow$
first higher level programming languages
relied on these ideas.